

**NATIONAL UNIVERSITY OF MODERN
LANGUAGE ISLAMABAD**

Lab 07



Submitted to:

Sir Hassan Shah

Submitted by:

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Roll No: SP21379

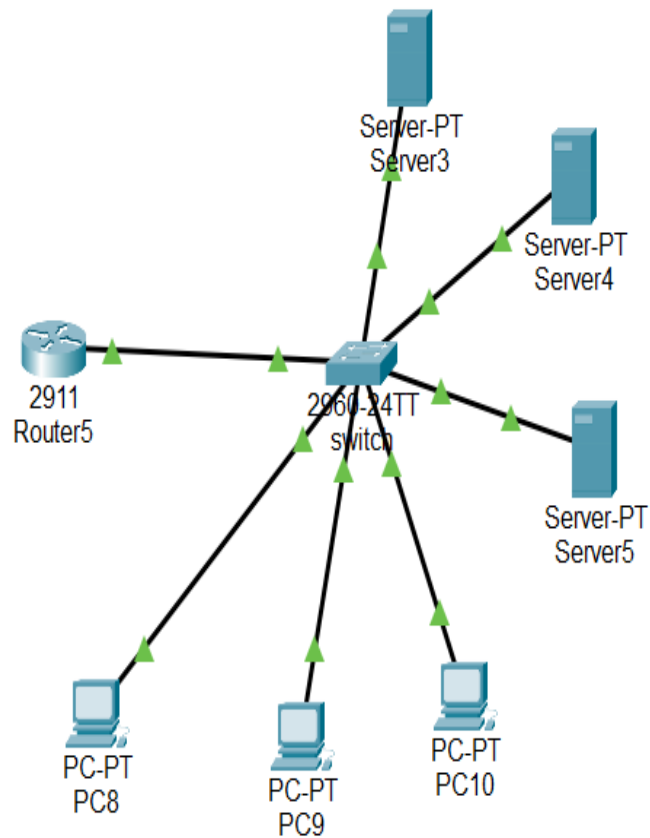
Subject: computer network

Department: BSSE-5-Afternoon

Date: 06-12-2024

Objectives:

- Using CDP protocol to find out neighbour information in Cisco packet tracer.
- Setting up a DHCP server in Cisco packet tracer.
- Configure a WEB server in the Cisco packet tracer and access the internet.
- Configuring DNS server.



Task 1

```
Zulqarnain-SP21379#
Zulqarnain-SP21379#show c
Zulqarnain-SP21379#show cd
Zulqarnain-SP21379#show cdp n
Zulqarnain-SP21379#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID      Local Intfcae  Holdtme  Capability  Platform  Port ID
Switch         Gig 0/0      149      S           2960      Gig 0/1
Router         Gig 0/1      172      R           C1900     Gig 0/1
Zulqarnain-SP21379#
Zulqarnain-SP21379#
Zulqarnain-SP21379#
Zulqarnain-SP21379#show cdp neighbors detail

Device ID: Switch
Entry address(es):
Platform: cisco 2960, Capabilities: Switch
Interface: GigabitEthernet0/0, Port ID (outgoing port): GigabitEthernet0/1
Holdtime: 126

Version :
Cisco IOS Software, C2960 Software (C2960-LANBASEK9-M), Version 15.0(2)SE4, RELEASE
SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2013 by Cisco Systems, Inc.
Compiled Wed 26-Jun-13 02:49 by mnguyen

advertisement version: 2
Duplex: full
-----

Device ID: Router
Entry address(es):
Platform: cisco C1900, Capabilities: Router
Interface: GigabitEthernet0/1, Port ID (outgoing port): GigabitEthernet0/1
Holdtime: 149

Version :|
Cisco IOS Software, C1900 Software (C1900-UNIVERSALK9-M), Version 15.1(4)M4, RELEASE
SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2012 by Cisco Systems, Inc.
Compiled Thurs 5-Jan-12 15:41 by pt_team

advertisement version: 2
Duplex: full

Zulqarnain-SP21379#
Zulqarnain-SP21379#
```

Task 3

```
Zulqarnain-SP21379(config)#interface g
Zulqarnain-SP21379(config)#interface gigabitEthernet 0/1
Zulqarnain-SP21379(config-if)#no c
Zulqarnain-SP21379(config-if)#no cd
Zulqarnain-SP21379(config-if)#no cdp e
Zulqarnain-SP21379(config-if)#no cdp enable
Zulqarnain-SP21379(config-if)#
Zulqarnain-SP21379(config-if)#
Zulqarnain-SP21379(config-if)#exit
Zulqarnain-SP21379(config)#exit
```

Task 4

```
Zulqarnain-SP21379#show cdp
Global CDP information:
  Sending CDP packets every 60 seconds
  Sending a holdtime value of 180 seconds
  Sending CDPv2 advertisements is enabled
Zulqarnain-SP21379#
Zulqarnain-SP21379#
Zulqarnain-SP21379#no cdp run
                        ^
% Invalid input detected at '^' marker.
Zulqarnain-SP21379#
```

DHCP

Task 5

Configure the router for DHCP on the given IP addresses and connect a switch and three PCs. To obtain IP addresses from DHCP, open the IP configuration on the PC. Attach a screenshot of the PCs receiving IP addresses through DHCP.

Configuring router

```
zulqarnain-sp21379(config)#ip dhcp pool My-networt
zulqarnain-sp21379(dhcp-config)#network 192.168.1.0 255.255.255.0
                        ^
% Invalid input detected at '^' marker.

zulqarnain-sp21379(dhcp-config)#network 192.168.1.0 255.255.255.0
zulqarnain-sp21379(dhcp-config)#default-router 192.168.1.1
zulqarnain-sp21379(dhcp-config)#exit
zulqarnain-sp21379(config)#
zulqarnain-sp21379(config)#ip dhcp ex
zulqarnain-sp21379(config)#ip dhcp excluded-address 192.168.1.1 192.168.1.10
zulqarnain-sp21379(config)#exit
zulqarnain-sp21379#
%SYS-5-CONFIG_I: Configured from console by console

zulqarnain-sp21379#
```

PC5

Physical Config Desktop Programming Attributes

IP Configuration

InterfaceFastEthernet0

IP Configuration

☒ DHCP

☐ Static

IPv4 Address169.254.55.147

Subnet Mask255.255.0.0

Default Gateway0.0.0.0

DNS Server0.0.0.0

IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address /

Link Local AddressFE80::230:F2FF:FE23:3792

Default GatewayDNS Server

802.1X

☐ Use 802.1X Security

AuthenticationMD5

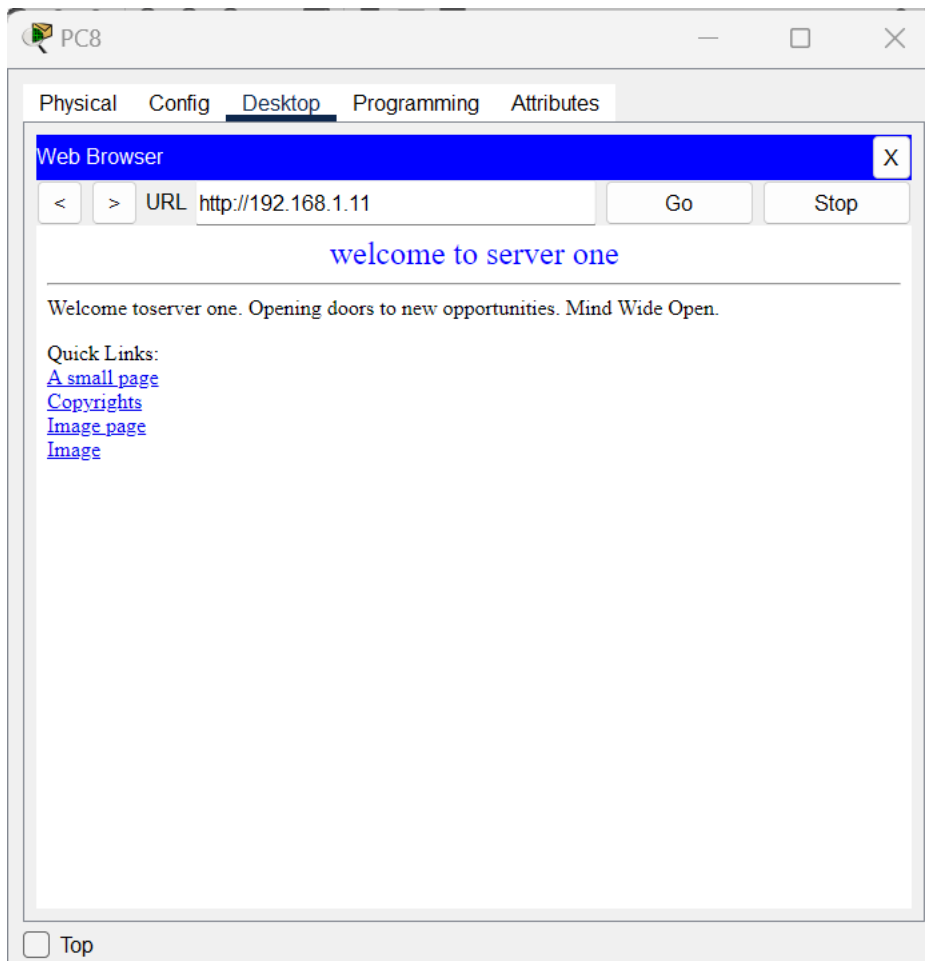
UsernamePassword

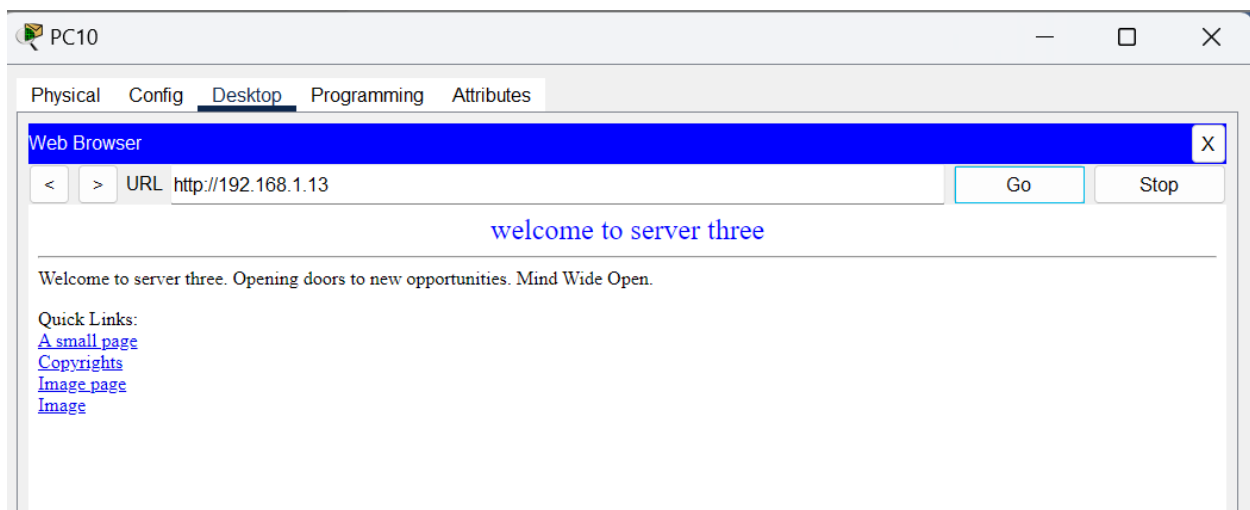
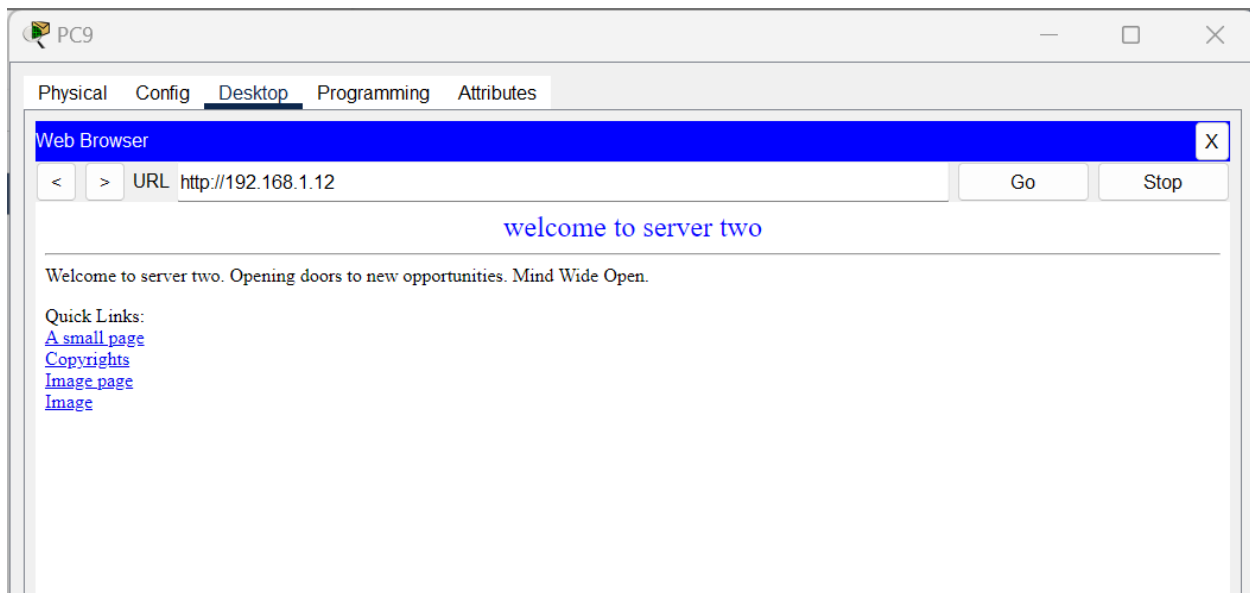
WEB SERVER

Task 6

Add different servers to other sites and access those sites on the PC browser. Attach a screenshot of the PCs accessing these sites.

```
zulqarnain-sp21379(config)#
zulqarnain-sp21379(config)#ip d
zulqarnain-sp21379(config)#ip dh
zulqarnain-sp21379(config)#ip dhcp pool My-network
zulqarnain-sp21379(dhcp-config)#net
zulqarnain-sp21379(dhcp-config)#network 192.168.1.0 255.255.255.0
zulqarnain-sp21379(dhcp-config)#de
zulqarnain-sp21379(dhcp-config)#default-router 192.168.1.1
zulqarnain-sp21379(dhcp-config)#dns
zulqarnain-sp21379(dhcp-config)#dns-server 192.168.1.1
zulqarnain-sp21379(dhcp-config)#exit
zulqarnain-sp21379(config)#ip dhcp ex
zulqarnain-sp21379(config)#ip dhcp excluded-address 192.168.1.1 192.168.1.10
zulqarnain-sp21379(config)#
zulqarnain-sp21379(config)#inter
zulqarnain-sp21379(config)#interface g
zulqarnain-sp21379(config)#interface gigabitEthernet 0/0
zulqarnain-sp21379(config-if)#ip address 192.168.1.1 255.255.255.0
zulqarnain-sp21379(config-if)#no sh
zulqarnain-sp21379(config-if)#no shutdown
zulqarnain-sp21379(config-if)#
zulqarnain-sp21379(config-if)#
```





DNS

Task 7

Create a mapping for the switch in the server so that PCs can ping the switch with the name 'switch'.

```

zulqarnain-sp21379(config)#inter
zulqarnain-sp21379(config)#interface vlan 1
zulqarnain-sp21379(config-if)#ip add
zulqarnain-sp21379(config-if)#ip address 192.168.1.2 255.255.255.0
zulqarnain-sp21379(config-if)#
zulqarnain-sp21379(config-if)#no
zulqarnain-sp21379(config-if)#no sh
zulqarnain-sp21379(config-if)#no shutdown
zulqarnain-sp21379(config-if)#exit
zulqarnain-sp21379(config)#
zulqarnain-sp21379(config)#

```

